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MAI–201

M. B. A. (AI & DS) (SECOND SEMESTER)

END SEMESTER EXAMINATION, June, 2023

BUSINESS ENVIRONMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question of Section A. Each sub-question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section–A

1. (a) Development of business society largely depends upon exploitation of natural resources. Keeping this fact in mind write a note on Technology-Nature Interdependence for balanced economic development.

(CO1, CO2, CO3)

(b) "Business has a deep social orientation." Critically examine the statement in the light of value, viability and visibility of business.

(CO1, CO2, CO3)

(c) Legal and government support environment plays a very important role in investment decision-making process. Explain it in the context of external factor affecting business environment. (CO1, CO2, CO3)

P. T. O.

2. (a) Write a brief note on *two* important internal factors affecting business decision-making. (CO2, CO3)

   (b) List important economic objectives of business. How these objectives reinforce the social objectives of business ? (CO2, CO3)

   (c) What do you understand by Business Ethics ? Elaborate on major ethical principles to be followed in business. (CO2, CO3)

3. (a) Write a note on corporate governance. What are the important factors affecting governance structure of a company ? (CO2, CO3, CO4)

   (b) What do you understand by CSR ? Discuss the A. B. Carroll's CSR Model. (CO2, CO3, CO4)

   (c) Explain the objectives, role and relevance of SEBI as a guardian of Indian financial market. (CO2, CO3, CO4)

4. (a) What do you understand by globalization ? How it has added to the cross border transmission of capital and technology ? (CO3, CO4, CO5)

   (b) What is monetary policy ? Discuss *two* important tools of monetary policy to control the inflationary pressure in the economy. (CO3, CO4, CO5)

   (c) Privatization of government enterprises was initiated with a motive to strengthen the structural reforms in the economy. What are the probable disadvantages of it ? (CO3, CO4, CO5)

Section—B

5. **Case Study :** (20 Marks)

In early 2002, Coca-Cola India (CCI) launched a new advertisement campaign featuring leading Bollywood actor–Aamir Khan. The advertisement with the tag line–'Thanda Matlab Coca-Cola' was targetted at rural and semi-urban consumers. According to company sources, the idea was to position Coca-Cola as a generic brand for cold drinks.

The campaign was launched to support CCI's rural marketing initiatives. CCI began focusing on the rural market in the early 2000s in order to increase volumes.

However, the poor rural infrastructure and consumption habits that are very different from those of urban people were two major obstacles to cracking the rural market for CCI.

Because of the erratic power supply most grocers in rural areas did not stock cold drinks. Also, people in rural areas had a preference for traditional cold beverages such as 'lassi'6 and lemon juice.

Further, the price of the beverage was also a major factor for the rural consumer.

Once CCI entered the rural market, it focused on strengthening its distribution network there. It realized that the centralized distribution system used by the company in the urban areas would not be suitable for rural areas.

In the centralized distribution system, the product was transported directly from the bottling plants to retailers. However, CCI realized that this distribution system would not work in rural markets, as taking stock directly from bottling plants to retail stores would be very costly due to the long distances to be covered.

The company instead opted for a hub and spoke distribution system. Under the hub and spoke distribution system, stock was transported from the bottling plants to hubs and then from hubs, the stock was transported to spokes which were situated in small towns. These spokes fed the retailers catering to the demand in rural areas. CCI not only changed its distribution model, it also changed the type of vehicles used for transportation. The company used large trucks for transporting stock from bottling plants to

hubs and medium commercial vehicles transported the stock from the hubs to spokes.

For transporting stock from spokes to village retailers the company utilized auto rickshaws and cycles. Commenting on the transportation of stock in rural markets, a company spokesperson said, "We use all possible means of transport that range from trucks, auto rickshaws, cycle rickshaws and hand carts to even camel carts in Rajasthan and mules in the hilly areas, to cart our products from the nearest hub."

In late 2002, CCI made an additional investment of ₹ 7 million (₹ 5 million from the company and ₹ 2 million from the company's bottlers) to meet rural demand. By March 2003, the company had added 25 production lines and doubled its glass and PET bottle capacity.

A survey conducted by CCI in 2001 revealed that 300 ml bottles were not popular with rural and semi-urban residents where two persons often shared a 300 ml bottle. It was also found that the price of ₹ 10 per bottle was considered too high by rural consumers.

The initiatives of CCI in distribution and pricing were supported by extensive marketing in the mass media as well as through outdoor advertising. The company put up hoardings in villages and painted the name Coca Cola on the compounds of the residences in the villages. Further, CCI also participated in the weekly mandies by setting up temporary retail outlets, and also took part in the annual haats and fairs–major sources of business activity and entertainment in rural India.

Question :

Is this a success story ? Comment on the strategy.

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MAI–203

M. B. A. (AI & DS) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023

HUMAN RESOURCE MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question of Section A. Each sub-question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section–A

1. (a) Analyze significance of Strategic Human Resource Management for IR 4.0. (CO3, CO4, CO5)

(b) Explain evolution of HRM from prominent management philosopher's perspective. (CO3, CO4, CO5)

(c) Write ten human resource management commandments from Indian scriptures. (CO3, CO4, CO5)

P. T. O.

2. (a) HRIS facilitates operational HR functions, efficiently. Analyze this statement with the help of an example. (CO2, CO3, CO4)

   (b) Evaluate role of different sources of E- recruitment in quality hiring. (CO3, CO4)

   (c) You are going to get hired by the Big 4 organization, explain different steps of selection process for your dream job profile. (CO4, CO5)

3. (a) Enumerate challenges of implementing 360 & 720 degree appraisal system in banking industry. (CO3, CO4, CO5)

   (b) Learning & Development budget is an investment for the High performing organization. Justify this sentence by taking an example of an L & D programme of a retailing firm. (CO3, CO4, CO5)

   (c) Develop a flow chart of your own career planning by mapping career goals and competence during different timeline/career stages. (CO3, CO4, CO5)

4. (a) How HR department can leverage HR analytics for organizational excellence. Explain. (CO3, CO4, CO5)

   (b) Technology is disrupting work, workplace & workforce. Explain challenges faced by an HR department, due to technological disruption. (CO3, CO4, CO5)

   (c) Analyze how this HR course will help you during delivering professional assignments. Enumertae ten practical implications of this course in your career. (CO4, CO5, CO6)

Section—B

5. **Caste Study** : (20 Marks) (CO4, CO5, CO6)

Testing Right Candidate for Right Job As companies try to streamline their hiring process and get the most appropriate candidate for the job, a two-year old startup is attempting to equip them with the right assessment tools. The Software As a Service (SaaS) company, Mettl, was founded by two school

friends Ketan Kapoor, an IIT Roorkee and IIM-Calcutta alumnus, and Tonmoy Shingal, an IIT-Kanpur and IIM-Bangalore alumnus. The software as a Service (SaaS) platform allows companies to create assessments based on parameters and skills required so thye can screen candidates for hiring. The platform can create assessments based on automatic evaluators and simulator. It can also create multi-competency assessments. "Our plan was to start a training and assessment company, but we quickly realised that assessment alone is a massive opportunity," says Kapoor. "Learning and assessments are moving online and that is why we created an online platform." As both the founders were not coders, they first built the technology team to create the platform. In late-2010, the company bagged its first client, Capgemini. "We kept improving the platform with inputs from clients. We have perfected it over two years," Kapoor says. The SaaS platform once installed can be run independently by the client. Using tools

provided within the platform, the recruiter can create assessments based on the parameters and skill sets required and the test can be taken by any number of test takers online. The recruiter gets assessment reports in real time. The Gurgaon-based company started off with a focus on the technology sector and is adding on modules for BPO, Banking and Education sector It now has over 50 customers, including Cognizant, Capgemini, InMobi, Ericsson, Cleartrip, Myntra and Yebhi. Mettl does have competition. Interviewstreet, an Indian startup incubated by US- based Y Combinator, creates coding contests to hire candidates for mostly US-based companies. Then there is Aspiring Minds, founded in 2007, which offers a computer-based assessment to test skills of students from engineering, MBA and other graduate institutes. However, Kapoor is unfazed. "Our platform is

highly scalable and can create tests across skill sets. We have a client who has created more than 50 types of assessments on the platform," adds Kapoor, who was earlier Business Development and Internet Marketing Manager for London-based online travel venture Isango. Capgemini's Vice-President, HR and India Talent Head, Sreenivasa Rao says he came across Mettl when the company was hunting for a technology platform that would help them streamline the assessment process. "We hire around 1,500 people every month and screen around 20,000 candidates to hire this number. Now, we use Mettl for 60% of our online assessments," says Rao. "We find the platform extremely adaptable and highly customizable. It fits our needs." Mettl later, it raised an angel round of investment from Blume Ventures, Sasha was launched with seed funding from friends and family in 2010. A year Mirchandani of Kae Capital, Naveen Tiwari of InMobi and other angels in 2011. Earlier this year, the company raised an institutional round of funding of $4 million from IndoUS Venture Partners and other existing investor Mirchandani exited during this round. Kapoor, who did not want to disclose revenues, says the funding will help the company provide more in-depth assessment modules in each sector it is already market, we want to launch operations in developing ad developed present in and also expand to other sectors. "Once we crack the India says Kapoor. He is planning to launch a training application for consumers that will be built atop the Mettl platform. *Source* : The Economic Times, 15th October 2012.

Questions :

(i) How effective are assessment tests in hiring the right candidate ?

(ii) In your opinion how is Mettl differentiating itself from its competitor.

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MAI–204

M. B. A. (AI & DS) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023

STATISTICS FOR BUSINESS DECISION

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) The median and mode of the following wage distribution are ₹ 33.5 and ₹ 343 respectively. However, three frequencies are missing. Determine their values : (CO1)

Wages (in hundred ₹)	Frequencies
0—10	4
10—20	16
20—30	?
30—40	?
40—50	?
50—60	6
60—70	4
Total	230

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(b) An analysis of production rejects resulted in the following figures :

(CO1)

No. of rejects per operator	No. of operator
21—25	5
26—30	15
31—35	28
36—40	42
41—45	15
46—50	12
51—55	3

Calculate the coefficient of skewness and kurtosis and comment on the results.

(c) From the following frequency distribution of heights of 360 boys in the age group 15-20 years, calculate the standard deviation and variance :

(CO1)

Height (in cms.)	No. of boys
126—130	31
131—135	44
136—140	48
141—145	51
146—150	60
151—155	55
156—160	43
161—165	28

2. (a) Solve the following system of Linear equation : (CO2)

$$x + 2y + z = 7; x + 3z = 11; 2x - 3y = 1$$

(b) The University of Graphic Era has given permission to four colleges in Dehradun city to start self financing three years B. A. (H) Economics course in 2023. The fee for each semester is fixed at ₹ 20 thousand per student. The average enrolment in six semesters in these four colleges are expected to be 50, 45, 40 and 35. The resource sharing ratio between the university and the colleges is 2 : 3. How much money the university of Graphic Era can expect to earn in the coming three years from these colleges ? (CO2)

(c) Evaluate : (CO2)

$$[x\, y\, z]\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

3. (a) A panel of two judges, A and B, graded seven T. V. serial performances by awarding marks independently as shown in the following table :

Performance	Marks by A	Marks by B
1	46	40
2	42	38
3	44	36
4	40	35
5	43	39
6	41	37
7	45	41

The eighth T. V. performance, which judge B, could not attend, was awarded 37 marks by judge A. If the judge B had also been present, how many marks would be expected to have been awarded by him to the eighth T. V. Performance ? (CO3)

(b) The following figures related to length of service and income of the employees of an organization :

Length of Services (Years)	Income (₹ Hundred)
11	7
7	5
2	3
5	2
8	6
6	4
10	8

Compute the coefficient of correlation for the above data. Find the two regression equations and examine the relationship. (CO3)

(c) Explain clearly the meaning of Time Series Analysis. Mention its important components. Explain these components with examples. (CO3)

4. (a) The price (in ₹) of a commodity during 1995-2000 is given below. Fit a parabola to this data. Estimate the price of commodity for the year 2003.

Year	Price
1995	100
1996	107
1997	128
1998	140
1999	181
2000	192

Also plot the actual and trend values on graph. (CO4)

(b) Below are given the figures of production (in million tones) of a cement factory :

Year	Production (in m tones)
1990	77
1992	88
1993	94
1994	85
1995	91
1996	98
1997	90

(i) Fit a straight-line trend and tabulate the trend values.

(ii) What is monthly increase in the production of cement ?

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(c) The sale of a commodity in tones varied from January, 2000 to December, 2000 in the following manner :

280, 300, 280, 280, 270, 240, 230, 230, 220, 200, 210, 200

Fit a trend line by the method of semi averages. Also plot the graph. (CO4)

5. (a) In a certain sample of 2000 families, 1400 families are consumers of tea, out of 1800 Hindu families, 1236 families consume tea. Test whether there is any significant difference between consumption of tea among Hindu and non-Hindu families. Tabulate value is 3.841. (CO5)

(b) Find the Eigen vector of the matrix : (CO5)

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}.$$

(c) Explain, why we need fuzzy set theory ? Differentiate between a fuzzy set and a crisp set. Find some examples of prospective fuzzy variable in daily life. (CO5)

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Roll No. .....

MAI–205

M. B. A. (AI & DS) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023

MACHINE LEARNING USING R

Time : Three Hours

Maximum Marks : 100

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *two* sub-questions among (a), (b) and (c) in each main question of Section A. Each sub-question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section–A

1. (a) What are support vectors, and how are they used to define the decision boundary in a support vector machine ? (CO1)

(b) Suppose you have a dataset of customer transactions, and you want to build a decision tree to predict whether or not a customer will make a purchase. The dataset has 1,000 records, with the following features : (CO1)

(i) Age (continuous variable)

(ii) Gender (binary variable: 0 for male, 1 for female)

(iii) Income (continuous variable)

(iv) Product category (categorical variable with 3 levels: A, B, and C)

(v) Purchase (binary variable : 0 for no purchase, 1 for purchase)

You decide to split the dataset into a training set with 800 records and a test set with 200 records. You also decide to use the Information Gain as the splitting criterion for the decision tree.

Here are the results of your decision tree model on the test set:

True positives : 50

False positives : 10

True negatives : 120

False negatives : 20

Using these results, calculate the following performance metrics for the decision tree model :

(i) Accuracy

(ii) Precision

(iii) Recall

(iv) F1-score

(c) Explain the difference between supervised and unsupervised learning. What are the advantages and disadvantages of each ? (CO1)

2. (a) How does logistic regression differ from linear regression, and what are the assumptions of logistic regression ? (CO4)

(b) What is cross-validation, and why is it used in machine learning ? Explain the difference between k-fold cross-validation and leave-one-out cross-validation. (CO3)

(c) What are some common techniques for combining the outputs of multiple models in an ensemble, and how do they work ? (CO4)

3. (a) Suppose you have a distance table for a set of 8 points, which you want to cluster using agglomerative clustering with single linkage. The table is given as follows : (CO2)

	Point 1	Point 2	Point 3	Point 4	Point 5	Point 6	Point 7	Point 8
Point 1	0.00	1.20	2.90	1.10	2.50	3.00	2.30	1.90
Point 2		0.00	3.30	1.40	3.20	3.10	2.60	2.10
Point 3			0.00	2.80	1.60	3.70	3.10	2.60
Point 4				0.00	2.90	2.90	1.40	0.90
Point 5					0.00	3.20	2.90	2.40
Point 6						0.00	4.30	3.90
Point 7							0.00	1.70
Point 8								0.00

(i) Using the distance table, perform agglomerative clustering with single linkage until there is only one cluster left. Show the dendrogram.

(ii) What is the final cluster, and which points belong to it?

Note : You can round your answers to two decimal places if necessary.

(b) What is the difference between classification and regression in machine learning ? Give an example of each. (CO2, CO5)

(c) Discuss the different evaluation metrics used in machine learning. How can you choose the appropriate metric for a given problem ? (CO2)

4. (a) What is reinforcement learning, and how is it different from supervised and unsupervised learning ? Give an example of an application of reinforcement learning. (CO5)

(b) Explain how R programming can be used in data analysis. Include a step-by-step guide on how to load and analyze a dataset using R, including data cleaning data exploration, and data visualization. (CO4)

(c) Describe the neural network architecture and how it is used in machine learning. What are the advantages and disadvantages of neural networks ? (CO5)

Section—B

(Case Study)

5. Read the following case study and answer the questions below :

A healthcare provider wants to improve patient outcomes by predicting which patients are at high risk of developing complications after surgery. The healthcare provider has a large database of patient medical records, including their pre-operative health status, medical history, and demographic information. The healthcare provider wants to use this data to build a

predictive analytics model that can identify high-risk patients and prioritize their care.

Questions : (CO5)

(a) What type of algorithm would you recommend for this case study and why ?

(b) What are some of the predictor variables that you would consider for this model ?

(c) What evaluation metrics would you use to assess the performance of this model ?

(d) How would you use the results of this model to make the decisions ?

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MAI–206

M. B. A. (AI AND DS) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023

DATA SCIENCE USING PYTHON

Time : Three Hours

Maximum Marks : 100

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *two* sub-questions among (a), (b) and (c) in each main question of Section A. Each sub-question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section–A

1. (a) Explain the process of data cleaning and preprocessing in Python. (CO2)

   (b) What is the difference between supervised and unsupervised learning in machine learning ? Give an example of each. (CO1)

   (c) How can you handle missing data in a dataset ? Describe different imputation techniques with examples. (CO3)

2. (a) Explain the concept of feature scaling in machine learning. Why is it important ? Describe different methods for feature scaling. (CO2)

   (b) What is EDA ? Explain the importance of EDA in Data Science. (CO4)

   (c) Explain the concept of outlier detection. How can you detect outliers in a dataset using Python ? (CO3)

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3. (a) Explain the concept of decision trees in machine learning. What are the advantages and disadvantages of decision trees ? (CO6)

   (b) What is logistic regression ? Explain the assumptions and limitations of logistic regression. (CO5)

   (c) What is clustering ? How is it performed in Python ? (CO4)

4. (a) What is deep learning ? How does it differ from traditional machine learning ? (CO2)

   (b) Explain the concept of Artificial Neural Networks (ANN) in deep learning. How can you implement an ANN in Python using Keras ? (CO4)

   (c) What is Convolutional Neural Networks (CNN) ? Explain the architecture of a CNN and its applications in image classification. (CO5)

Section–B

5. **Case Study :** (20 Marks) (CO6)

Customer Churn Prediction

A telecommunications company wants to predict customer churn so that they can take proactive measures to retain customer. The company has collected data on customer behavior, demographics and account information. They have hired a data scientist to build a predictive model to indentify customers who are likely to churn.

Question :

Build a suitable model and provide steps for above problem assuming dataset.